

NONPARAMETRIC REGRESSION FOR CONSECUTIVE SURVIVAL DATA

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Summary: In statistical applications to medical data one often encounters situations when one is interested in sequentially observed periods of a disease. Relevant data may then be heavily truncated and/or censored. In this paper we extend the famous Nadaraya-Watson estimator of regression to such data and study its distributional behavior.

1. Introduction and Main Result

In medical studies the total observational period of a patient is often split into several consecutive periods describing various levels in the development of a disease. For example, in retrospective AIDS studies, the first period may start at the time of infection and last until seroconversion, while the second phase may start then and end at the calendar time of diagnosis of AIDS. Let $X_1 \leq X_2 \leq X_3$ denote the three relevant time points, and denote with

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$U_1 = X_2 - X_1$ and $U_2 = X_3 - X_2$ the lengths of the associated periods. One may then be interested, e.g., in the (joint) distributional behavior of the U 's or, alternatively, in the prediction of U_2 given that only the incubation period U_1 is available. Unfortunately, the U 's are often not fully observable. Typically, in a medical context, cases enter the study only when $X_2 \leq E$, the day when the observational period ends. Hence, if we put $Z = E - X_1$, we observe U_1 only when $U_1 \leq Z$. This so-called truncation effect naturally needs to be respected in the data analysis. Furthermore, we may have censoring effects in that we observe U_2 only if $U_1 + U_2 \leq Z$ resp. $X_3 \leq E$ and $Z - U_1$ otherwise. Summarizing, given $U_1 \leq Z$, we have $U_1, \tilde{U}_2 \equiv \min\{U_2, Z - U_1\}$ and $\delta = 1_{\{U_1 + U_2 \leq Z\}}$, indicating whether U_2 or some censoring variable was observed. Let $(U_{1i}, \tilde{U}_{2i}, \delta_i), 1 \leq i \leq n$, be a given sample of independent replicates of $(U_1, \tilde{U}_2, \delta)$. The goal then is to recover distributional properties of (U_1, U_2) from the given data. To introduce some notation, let

$$F(u_1, u_2) = \mathbb{P}(U_1 \leq u_1, U_2 \leq u_2)$$

be the unknown joint distribution function (d.f.) of U_1 and U_2 . We also need the marginals

$$F_1(u_1) = \mathbb{P}(U_1 \leq u_1) \text{ and } G(x) = \mathbb{P}(Z \leq x).$$

Since we only observe the U_1 's which were not truncated, the probability

$$1 - \alpha = \mathbb{P}(U_1 > Z)$$

of truncation is also important. We may think of $U_{1i}, 1 \leq i \leq n$, as coming from a larger sample of N data $U_{1i}, 1 \leq i \leq N$, such that

$$n = \sum_{i=1}^N 1_{\{U_{1i} \leq Z_i\}}. \quad (1.1)$$

The sample size N is unknown so that n/N is not available as an estimator of α . W.l.o.g. we may also assume that the first n of the U_{1i} 's have passed their thresholds. Typically, the statistical analysis of truncated data assumes that n is given.

Functions which are easily estimated are the conditional (sub-) distributions

$$F_1^*(x) = \mathbb{P}(U_1 \leq x | U_1 \leq Z) \quad G^*(x) = \mathbb{P}(Z \leq x | U_1 \leq Z)$$

and

$$H_2^1(x, y) = \mathbb{P}(U_1 \leq x, \tilde{U}_2 \leq y, \delta = 1 | U_1 \leq Z).$$

Throughout the paper we assume that the original vector

$$(A1) \quad (U_1, U_2) \text{ is independent of } Z.$$

It follows that for $x, y \geq 0$

$$H_2^1(x, y) = \alpha^{-1} \int_{[0, x]} \int_{[0, y]} [1 - G^-(x_1 + x_2)] F(dx_1, dx_2).$$

Here G^- is the left-continuous version of G . For each function φ which vanishes on the set of (x_1, x_2) with $G^-(x_1 + x_2) = 1$, we obtain

$$I \equiv \int \varphi dF = \int \varphi(x_1, x_2) \frac{\alpha}{1 - G^-(x_1 + x_2)} H_2^1(dx_1, dx_2).$$

Finally, again by independence,

$$F_1^*(x) = \alpha^{-1} \int_{[0, x]} (1 - G^-(u)) F_1(du)$$

and

$$G^*(y) = \alpha^{-1} \int_{[0, y]} F_1(z) G(dz).$$

Now, to estimate F_1 and G from a truncated sample, we need some constraints on their support. For any d.f. D , set

$$a_D = \inf\{x \in \mathbb{R} : D(x) > 0\}$$

and

$$b_D = \inf\{x \in \mathbb{R} : D(x) = 1\} \quad (\leq \infty).$$

Then, under

$$(A2) \quad b_{F_1} \leq b_G \text{ and } a_{F_1} \leq a_G$$

we obtain

$$\begin{aligned} A(x) &\equiv \alpha^{-1}(1 - G^-(x)) = \alpha^{-1} \int_{[x, \infty)} G(dy) \\ &= \int_{[x, \infty)} \frac{G^*(dy)}{F_1(y)} \end{aligned}$$

and thus

$$I = \int \varphi dF = \int \varphi(x_1, x_2) \frac{1}{A(x_1 + x_2)} H_2^1(dx_1, dx_2).$$

It becomes clear that the function A plays a crucial role in our analysis. The nonparametric product limit estimator of F_1 is the Lynden-Bell (1971) estimator given by

$$F_{1n}(t) = \prod_{U_{1i} > t} \left[1 - \frac{1}{nC_n(U_{1i})} \right],$$

with

$$C_n(x) = \frac{1}{n} \sum_{i=1}^n 1_{\{U_{1i} \leq x \leq Z_i\}}$$

being the average number of pairs (U_{1i}, Z_i) covering a point x . If $nC_n(U_{1i}) = 1$ for some $1 \leq i \leq n$, then U_{1i} is only covered by (U_{1i}, Z_i) so that F_{1n} vanishes at all points strictly less than U_{1i} . Strzalkowska-Kominiak and Stute (2010a) provided a detailed analysis of the probability of such holes. Among other things it could be shown that on an event Ω_n whose probability tends to one as $n \rightarrow \infty$, we have that

$$T_n := \max\{U_{1i} : nC_n(U_{1i}) = 1\}$$

equals $\min_{1 \leq i \leq n} U_{1i}$. Hence, if we set

$$A_n(x) = \int_{[x, \infty)} \frac{G_n^*(dy)}{F_{1n}(y)}$$

with

$$G_n^*(y) = \frac{1}{n} \sum_{i=1}^n 1_{\{Z_i \leq y\}},$$

then $A_n(x)$ is well-defined on Ω_n .

It remains to recall H_2^1 and define its nonparametric estimator through

$$H_{2n}^1(u, v) = n^{-1} \sum_{i=1}^n 1_{\{U_{1i} \leq u, U_{2i} \leq v, \delta_i = 1\}}.$$

We are now in the position to construct, for a given φ , our estimator of $I = \int \varphi dF$:

$$\begin{aligned} I_n &= n^{-1} \sum_{i=1}^n \varphi(U_{1i}, U_{2i}) \frac{\delta_i 1_{\{U_{1i} + U_{2i} \geq T_n\}}}{A_n(U_{1i} + U_{2i})} \\ &= \int \varphi(u_1, u_2) \frac{1_{\{u_1 + u_2 \geq T_n\}}}{A_n(u_1 + u_2)} H_{2n}^1(du_1, du_2). \end{aligned}$$

In Strzalkowska-Kominiak and Stute (2010b), we derived an i.i.d. representation for I_n and showed its asymptotic normality. If for φ we take $\varphi(u, v) = 1_{\{u \leq u_1, v \leq u_2\}}$, I_n becomes an estimator of $F(u_1, u_2)$ denoted with $F_n(u_1, u_2)$:

$$F_n(u_1, u_2) = n^{-1} \sum_{i=1}^n 1_{\{U_{1i} \leq u_1, U_{2i} \leq u_2\}} \frac{\delta_i 1_{\{U_{1i} + U_{2i} \geq T_n\}}}{A_n(U_{1i} + U_{2i})}. \quad (1.2)$$

Note that F_n is a “bona fide” distribution function and boils down to the classical empirical d.f. of the (U_{1i}, U_{2i}) if $Z_i = \infty$, i.e., when we have no truncation and censorship.

In the present paper we are not interested in the (joint) distributional behavior of (U_1, U_2) but in the regression of U_2 on U_1 :

$$m(x) = \mathbb{E}[U_2 | U_1 = x].$$

In a completely nonparametric framework Nadaraya (1964) and Watson (1964) proposed an estimator of m obtained from local kernel smoothing. For this, let K be a nonnegative square-integrable function. Suppose that K is odd so that $\int xK(x)dx = 0$. W.l.o.g. we may also assume that $\int K(x)dx = 1$. Fix a positive bandwidth h and set

$$K_h(u) = \frac{1}{h} K\left(\frac{u}{h}\right).$$

Assume for a moment that there is no truncation and censorship. The NW-estimator equals

$$m_n(x) = \frac{\sum_{i=1}^n U_{2i} K_h(x - U_{1i})}{\sum_{i=1}^n K_h(x - U_{1i})} \equiv \sum_{i=1}^n W_{ni}(x) U_{2i} \quad (1.3)$$

with

$$W_{ni}(x) = \frac{K_h(x - U_{1i})}{\sum_{j=1}^n K_h(x - U_{1j})}.$$

Alternatively, in terms of the classical empirical d.f. F_n :

$$m_n(x) = \frac{\int u_2 K_h(x - u_1) F_n(du_1, du_2)}{\int K_h(x - u_1) F_n(du_1, du_2)}.$$

Under truncation and censorship, we may define m_n in the same way, but with F_n replaced by our extension from (1.2), i.e.,

$$m_n(x) = \frac{\int u_2 K_h(x - u_1) \frac{1_{\{u_1+u_2 \geq T_n\}}}{A_n(u_1+u_2)} H_{2n}^1(du_1, du_2)}{\int K_h(x - u_1) \frac{1_{\{u_1+u_2 \geq T_n\}}}{A_n(u_1+u_2)} H_{2n}^1(du_1, du_2)} \quad (1.4)$$

The main purpose of this paper is to study the distributional behavior of this $m_n(x)$ at a fixed x . This will be the content of Theorem 1.1. In Section 2 we present some simulation results while proofs are given in Section 3.

Apart from (A1) and (A2) the following regularity conditions are needed:

- (A3) • F admits a density f with a first marginal f_1 which is assumed to be twice continuously differentiable in a neighborhood of x .
- m is twice continuously differentiable in a neighborhood of x .
 - K is a bounded kernel satisfying $\int uK(u)du = 0$ and $\int u^2K(u)du < \infty$.
 - $\int \frac{|v|}{[1 - G^-(u + v)]^2} F(du, dv) < \infty$.

The first three assumptions are standard and the same as in the situation when no truncation or censorship is present. The last assumption puts a bound on the censoring effect guaranteeing that we have the same rate of convergence as for completely observable data.

We collect some identities which are important in the future. Denote with f^* and f_1^* the joint respectively first marginal density of (U_1, U_2) given $U_1 \leq Z$. Then, by (A1),

$$f^*(u_1, u_2) = A(u_1)f(u_1, u_2)$$

and

$$f_1^*(u_1) = A(u_1)f_1(u_1).$$

It follows that without censorship truncation has no effect on regression, i.e., U_2 admits the same regression on U_1 before and after truncation. Hence we could apply the classical NW-estimator to the available data if censorship is not present. As we shall see, things are different if in addition censoring effects occur.

Theorem 1.1 *Under (A1) – (A3) we have, as $h \rightarrow 0$ and $nh \rightarrow \infty$:*

$$(nh)^{1/2}\{m_n(x) - m(x)\} = (nh)^{1/2}(I + II),$$

where

$$(nh)^{1/2}I \rightarrow \mathcal{N}(0, \rho^2(x)) \text{ in distribution}$$

with

$$\rho^2(x) = \frac{\int \frac{(v-m(x))^2}{A(x+v)} f(x, v) dv \int K^2(u) du}{f_1^2(x)}.$$

Moreover, the bias term Π admits the expansion

$$\Pi = h^2 B(x) + o(h^2)$$

with

$$B(x) = \frac{\{2f_1'(x)m'(x) + f_1(x)m''(x)\} \int u^2 K(u) du}{2f_1(x)}.$$

It follows from Theorem 1.1 that the asymptotic Mean Square Error becomes

$$\text{AMSE} = \text{AMSE}(x, h) = h^4 B^2(x) + \frac{\rho^2(x)}{nh}.$$

Minimization w.r.t. h leads to

$$h_{\text{opt}} = cn^{-1/5} \text{ with } c = \left[\frac{\rho^2(x)}{4B^2(x)} \right]^{1/5}. \quad (1.5)$$

For other c 's we have the following consequence of Theorem 1.1.

Corollary 1.1 *Assume that $nh^5 \rightarrow c$ as $n \rightarrow \infty$. Then*

$$(nh)^{1/2} \{m_n(x) - m(x)\} \rightarrow \mathcal{N}(c^{1/2} B(x), \rho^2(x))$$

in distribution as $n \rightarrow \infty$.

Remark 1.1. The formulas for $\rho^2(x)$ and $B(x)$ should be compared with those in the classical case. Formally, this is achieved when we set $Z_i = \infty$. In this case $\alpha = 1$ and $G(x) = 0$ for all $x \in \mathbb{R}$ but $G(\infty) = 1$. Hence $A \equiv 1$ on the real line so that ρ^2 boils down to

$$\rho^2(x) = \frac{\int (v - m(x))^2 f(x, v) dv \int K^2(u) du}{f_1^2(x)}$$

or, alternatively, with

$$\sigma^2(x) = \text{Var}(U_2|U_1 = x)$$

denoting the conditional variance of U_2 given $U_1 = x$:

$$\rho^2(x) = \frac{\sigma^2(x) \int K^2(u) du}{f_1(x)}$$

Under truncation and censoring, the denominator $A(x+v)$ is less than or equal to one and hence increases the variance of our estimator when truncation and censorship are present. Moreover, the bias is the same as in the classical case.

See, e.g., Wand and Jones (1995).

Remark 1.2 We only mention that, at the cost of some extra notation, the results of our paper can be extended to a situation when, as before, X_2 may be truncated by some E_1 but X_3 is censored by E_2 , with $E_1 < E_2$. Hence the critical point E is replaced by some period $[E_1, E_2]$.

2. Simulation Results

In this section we investigate the performance of our estimator in various scenarios. They differ in the choice of m and in the percentages of truncation and censorship. We start with the description of the models. Denote with $\exp(b)$ the exponential distribution with parameter b . Also $U[a, b]$ denotes the uniform distribution over $[a, b]$.

MODEL 1 We have

$$U_1 \sim \exp(1), Z \sim \exp(b)$$

and

$$U_2 = 1 + U_1 + \varepsilon,$$

where $\varepsilon \sim U[-1, 1]$ is independent of U_1 . Hence $m(x) = 1 + x$. Furthermore, for $b = 0.1$ the truncation rate is 9% and the additional censoring rate equals

16%. For $b = 0.2$ the truncation rate is 17% and the additional censoring rate 29%.

MODEL 2

$$U_1 \sim \exp(1), Z \sim \exp(b)$$

and

$$U_2 = 2 + 0.2U_1 - e^{-(U_1-1)^2} + \varepsilon,$$

where $\varepsilon \sim U[-0.5, 0.5]$. Hence $m(x) = 2 + 0.2x - e^{-(x-1)^2}$. For $b = 0.1$ the truncation rate is approximately 9% and the additional censoring rate 14%. For $b = 0.2$ the truncation rate is 17% and the additional censoring rate 26%.

In the following tables, we compare sample bias, variance and mean square errors of $m_n(x)$ with bandwidth $\hat{h}$ and h_{opt} in Models 1 and 2. For $\hat{h}$, we choose

$$\hat{h} = \hat{c}n^{-1/5}$$

with $\hat{c}$ denoting a pivotal estimator of the optimal c from (1.5). The kernel K was always the standard normal kernel. The number of repetitions in the simulations was always $M = 500$.

Table 1. Model 1: MSE of $m_n(\hat{h})$ and $m_n(h_{\text{opt}})$

	$b = 0.1$			
	$x = 0.5$		$x = 1$	
	$N = 100$	$N = 200$	$N = 100$	$N = 200$
$MSE(m_n(\hat{h}))$	0.0131	0.0087	0.0287	0.0179
$MSE(m_n(h_{\text{opt}}))$	0.0119	0.0066	0.0184	0.0109
	$b = 0.2$			
	$x = 0.5$		$x = 1$	
	$N = 100$	$N = 200$	$N = 100$	$N = 200$
$MSE(m_n(\hat{h}))$	0.0157	0.0095	0.0352	0.0215
$MSE(m_n(h_{\text{opt}}))$	0.0132	0.0074	0.0219	0.0126

Table 2. Model 2: MSE of $m_n(\hat{h})$ and $m_n(h_{\text{opt}})$

	$b = 0.1$			
	$x = 0.5$		$x = 0.9$	
	$N = 100$	$N = 200$	$N = 100$	$N = 200$
$MSE(m_n(\hat{h}))$	0.0060	0.0040	0.0208	0.0136
$MSE(m_n(h_{\text{opt}}))$	0.0037	0.0021	0.0046	0.0027
	$b = 0.2$			
	$x = 0.5$		$x = 0.9$	
	$N = 100$	$N = 200$	$N = 100$	$N = 200$
$MSE(m_n(\hat{h}))$	0.0065	0.0038	0.0185	0.0103
$MSE(m_n(h_{\text{opt}}))$	0.0044	0.0025	0.0056	0.0030

It becomes clear that the estimator with a pivotal $\hat{c}$ is only slightly worse than m_n with the (asymptotically) optimal c .

3. Proofs

Recall $m_n(x)$ from (1.4). In the following we write $m_n(x) = N_n(x)/D_n(x)$. In our first lemma we compare the numerator $N_n(x)$ and the denominator $D_n(x)$ with the corresponding quantities $N_n^0(x)$ and $D_n^0(x)$ obtained after replacing A_n by A in (1.4). For this, put

$$\varphi_1(u, v) = K_h(x - u)v$$

and

$$\varphi_2(u, v) = K_h(x - u).$$

Then

$$N_n^0(x) = \int \frac{\varphi_1(u, v)1_{\{u+v \geq T_n\}}}{A(u+v)} H_{2n}^1(du, dv)$$

$$D_n^0(x) = \int \frac{\varphi_2(u, v)1_{\{u+v \geq T_n\}}}{A(u+v)} H_{2n}^1(du, dv).$$

Lemma 3.1 *Suppose that $h \rightarrow 0$ and $nh \rightarrow \infty$. Then, in probability,*

$$(nh)^{1/2}[N_n(x) - N_n^0(x)] \rightarrow 0 \quad (3.1)$$

and

$$(nh)^{1/2}[D_n(x) - D_n^0(x)] \rightarrow 0. \quad (3.2)$$

Proof. We already mentioned that, on a set Ω_n whose probability tends to one as $n \rightarrow \infty$, $T_n = \min_{1 \leq i \leq n} U_{1i}$. In particular, on this set,

$$1_{\{U_{1i} + U_{2i} \geq T_n\}} = 1.$$

From Lemma 4.3 in Strzalkowska-Kominiak and Stute (2010b) we know that the process A/A_n is bounded from above with probability tending to one. Furthermore, the process $n^{1/2}(A_n - A)$ is bounded in probability. Finally, to prove the lemma, note that

$$\frac{1}{A_n} - \frac{1}{A} = \frac{A - A_n}{A^2} + \frac{(A - A_n)^2}{A^2 A_n}.$$

On the set Ω_n , we therefore have

$$(nh)^{1/2}[N_n(x) - N_n^0(x)] = I_1 + I_2,$$

with

$$I_1 = (nh)^{1/2} \int \frac{\varphi_1(u, v)}{A^2(u + v)} (A - A_n)(u + v) H_{2n}^1(du, dv)$$

and

$$I_2 = (nh)^{1/2} \int \frac{\varphi_1(u, v)}{A^2 A_n(u + v)} (A - A_n)^2(u + v) H_{2n}^1(du, dv).$$

To show that $I_2 \rightarrow 0$ in probability, we use the above mentioned properties of A_n and the boundedness of K . By this it remains to prove that

$$(nh)^{-1/2} \int \frac{|v|}{A^3(u + v)} H_{2n}^1(du, dv)$$

tends to zero. But this holds, since $nh \rightarrow \infty$ and the integral tends to

$$\int \frac{|v|}{A^3(u + v)} H_2^1(du, dv) = \int |v| A^{-2}(u + v) F(du, dv) < \infty,$$

by the SLLN and Assumption (A3).

Next we deal with I_1 . From Lemma 4.2 in Strzalkowska-Kominiak and Stute (2010b) we know that

$$(A_n - A)(w) = \frac{1}{n} \sum_{j=1}^n T(w, U_{1j}, Z_j) + o_{\mathbb{P}}(n^{-1/2}).$$

where

$$\mathbb{E}T(w, U_1, Z_1) = 0 \text{ for all } w. \quad (3.3)$$

Conclude that, up to an $o_{\mathbb{P}}(1)$ term,

$$\begin{aligned} I_1 &= -(nh)^{1/2} \frac{1}{hn^2} \sum_{j=1}^n \sum_{i \neq j} K\left(\frac{x - U_{1i}}{h}\right) U_{2i} \delta_i \frac{T(U_{1i} + U_{2i}, U_{1j}, Z_j)}{A^2(U_{1i} + U_{2i})} \\ &\equiv -n^{-3/2} \sum_{j=1}^n \sum_{i \neq j} H_{ij}, \end{aligned}$$

where

$$H_{ij} = h^{-1/2} K\left(\frac{x - U_{1i}}{h}\right) U_{2i} \delta_i \frac{T(U_{1i} + U_{2i}, U_{1j}, Z_j)}{A^2(U_{1i} + U_{2i})}.$$

Finally, put

$$\bar{H}_j = \mathbb{E}(H_{ij} | U_{1j}, Z_j)$$

so that (up to a remainder)

$$I_1 = -n^{-3/2} \sum_{j=1}^n \sum_{i \neq j} (H_{ij} - \bar{H}_j) - \frac{n-1}{n} n^{-1/2} \sum_{j=1}^n \bar{H}_j. \quad (3.4)$$

Since by (3.3)

$$\mathbb{E}T(w, U_{1j}, Z_j) = 0 \text{ for all } w,$$

it follows that, for $i \neq j$,

$$\mathbb{E}(H_{ij} - \bar{H}_j | U_{1i}, U_{2i}, \delta_i) = 0$$

and

$$\mathbb{E}(H_{ij} - \bar{H}_j | U_{1j}, Z_j) = 0.$$

It follows that the first term in (3.4) is a degenerate U -statistic with expectation zero whose variance equals

$$\frac{n(n-1)}{n^3} [\mathbb{E}(H_{12} - \bar{H}_2)(H_{21} - \bar{H}_1) + \mathbb{E}(H_{12} - \bar{H}_2)^2]. \quad (3.5)$$

Since K is bounded and the remaining terms in the H_{ij} 's are square-integrable,

$$|(3.5)| = O((nh)^{-1}) = o(1).$$

It remains to study

$$n^{-1/2} \sum_{j=1}^n \bar{H}_j,$$

a sum of centered i.i.d. random variables with the same variance as $\bar{H}_1$. But

$$\begin{aligned} \bar{H}_1 &= h^{-1/2} \int K\left(\frac{x-u}{h}\right) v \frac{T(u+v, U_{11}, Z_1)}{A^2(u+v)} H_2^1(du, dv) \\ &= h^{-1/2} \int K\left(\frac{x-u}{h}\right) v \frac{T(u+v, U_{11}, Z_1)}{A(u+v)} f(u, v) du dv \\ &= h^{1/2} \int K(w) v \frac{T(x-hw+v, U_{11}, Z_1)}{A(x-hw+v)} f(x-wh, v) dw dv. \end{aligned}$$

Hence

$$\text{Var}(\bar{H}_1) = h \int \left[\int K(w) v \frac{T(x-hw+v, u, z)}{A(x-hw+v)} f(x-wh, v) dw dv \right]^2 H_2(du, dz),$$

where

$$H_2(u, z) = \mathbb{P}(U_1 \leq u, Z \leq z | U_1 \leq Z).$$

Since, as $h \rightarrow 0$, the integral converges, we obtain

$$\text{Var}(\bar{H}_1) = O(h) = o(1).$$

This completes the proof of (3.1). The arguments for (3.2) are similar and therefore omitted. ■

If we ignore the (asymptotically negligible) indicator $1_{\{u+v \geq T_n\}}$ for a moment, then $N_n^0(x)$ becomes the sample mean of the terms

$$h^{-1} \frac{K\left(\frac{x-U_{1i}}{h}\right) U_{2i} 1_{\{U_{1i}+U_{2i} \leq Z_i\}}}{A(U_{1i} + U_{2i})}.$$

Similarly for D_n^0 , in which case we only have to skip the factor U_{2i} . To understand the behavior of N_n^0 and D_n^0 , we compute the relevant regression functions.

Lemma 3.2 *We have, on the set $U_1 \leq Z$,*

$$\mathbb{E} \left[\frac{U_2 1_{\{U_1+U_2 \leq Z\}}}{A(U_1 + U_2)} \middle| U_1 = u, U_1 \leq Z \right] = \frac{m(u)}{A(u)} = m_1(u)$$

and

$$\mathbb{E} \left[\frac{1_{\{U_1+U_2 \leq Z\}}}{A(U_1 + U_2)} \middle| U_1 = u, U_1 \leq Z \right] = \frac{1}{A(u)} = m_2(u)$$

Proof. We only show the first equation, since the second can be handled in a similar way. First, by independence of (U_1, U_2) and Z , we obtain

$$\begin{aligned} & \mathbb{E} \left[\frac{U_2 1_{\{U_1+U_2 \leq Z\}}}{A(U_1 + U_2)} \middle| U_1, U_2, U_1 \leq Z \right] \\ &= \frac{U_2}{A(U_1 + U_2)} \mathbb{E} [1_{\{U_1+U_2 \leq Z\}} | U_1, U_2, U_1 \leq Z] \\ &= \frac{U_2}{A(U_1 + U_2)} \frac{1 - G^-(U_1 + U_2)}{1 - G^-(U_1)} = \frac{U_2}{A(U_1)}. \end{aligned}$$

Now condition on U_1 and $U_1 \leq Z$ and use $\mathbb{E}[U_2 | U_1] = m(U_1)$ to get the result. ■

The following lemma shows that to analyze m_n , it suffices to consider

$$m_n^0(x) = \frac{N_n^0(x)}{D_n^0(x)}.$$

Lemma 3.3 *We have, under (A1) – (A3),*

$$(nh)^{1/2} [m_n(x) - m_n^0(x)] \rightarrow 0 \text{ in probability.}$$

Proof. We have

$$\frac{N_n(x)}{D_n(x)} - \frac{N_n^0(x)}{D_n^0(x)} = \frac{N_n(x) - N_n^0(x)}{D_n(x)} + \frac{N_n^0(x)[D_n^0(x) - D_n(x)]}{D_n(x)D_n^0(x)}.$$

Introduce

$$f_{1n}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - U_{1i}}{h}\right),$$

the classical kernel density estimator for $f_1^*(x)$. Then apart from the negligible indicators $1_{\{u+v \geq T_n\}}$, $N_n^0(x)/f_{1n}(x)$ and $D_n^0(x)/f_{1n}(x)$ are standard Nadaraya-Watson estimators for $m_1(x)$ and $m_2(x)$, respectively.

Conclude that in probability (recall Lemma 3.1 for D_n and N_n)

$$\begin{aligned} \lim_{n \rightarrow \infty} D_n^0(x) &= \lim_{n \rightarrow \infty} D_n(x) = m_2(x)f_1^*(x) \\ \lim_{n \rightarrow \infty} N_n^0(x) &= \lim_{n \rightarrow \infty} N_n(x) = m_1(x)f_1^*(x). \end{aligned}$$

Next apply Lemma 3.1 to show that

$$m_n(x) - m_n^0(x) = o_{\mathbb{P}}((nh)^{-1/2}). \quad \blacksquare$$

Proof of Theorem 1.1 As a consequence of the last lemma it suffices to study m_n^0 . For this, set

$$m_{1n} = N_n^0/f_{1n} \text{ and } m_{2n} = D_n^0/f_{1n}$$

and recall $m = m_1/m_2$ from Lemma 3.2. Simple algebra yields

$$m_n^0 - m = \frac{m_{1n}m_2 - m_{2n}m_1}{m_{2n}m_2}. \quad (3.6)$$

The denominator converges to m_2^2 in probability. So we have everything boiled down to the analysis of two ordinary Nadaraya-Watson estimators, with the same inputs U_{1i} , $1 \leq i \leq n$, but different outputs

$$Y_{1i} = \frac{U_{2i}1_{\{U_{1i}+U_{2i} \leq Z_i\}}}{A(U_{1i} + U_{2i})}$$

and

$$Y_{2i} = \frac{1_{\{U_{1i}+U_{2i} \leq Z_i\}}}{A(U_{1i} + U_{2i})}.$$

Consequently the weights $W_{ni}(x)$, see (1.3), are the same for both m_{1n} and m_{2n} . It follows that the numerator in (3.6) admits the decomposition

$$\begin{aligned} & \sum_{i=1}^n W_{ni}(x) [Y_{1i}m_2(x) - Y_{2i}m_1(x)] \\ &= \sum_{i=1}^n W_{ni}(x) [(Y_{1i} - m_1(U_{1i}))m_2(x) - (Y_{2i} - m_2(U_{1i}))m_1(x)] \\ &+ \sum_{i=1}^n W_{ni}(x)[m_1(U_{1i}) - m_1(x)]m_2(x) \end{aligned} \quad (3.7)$$

$$- \sum_{i=1}^n W_{ni}(x)[m_2(U_{1i}) - m_2(x)]m_1(x). \quad (3.8)$$

The first sum is a sum of a martingale difference array which, after multiplication with $(nh)^{1/2}$, has a normal limit. Just apply results from Brown and Eagleson (1971). Its predictable quadratic variation equals

$$\begin{aligned} nh \sum_{i=1}^n W_{ni}^2(x) \rho_1^2(U_{1i}) &= \frac{\frac{1}{nh} \sum_{i=1}^n K^2\left(\frac{x-U_{1i}}{h}\right) \rho_1^2(U_{1i})}{f_{1n}^2(x)} \\ &\rightarrow \frac{\rho_1^2(x) \int K^2(u) du}{f_1^*(x)}, \end{aligned} \quad (3.9)$$

with pertaining conditional variance

$$\rho_1^2(x) = \mathbb{E} \left\{ [(Y_{1i} - m_1(U_{1i}))m_2(x) - (Y_{2i} - m_2(U_{1i}))m_1(x)]^2 \mid U_{1i} = x, U_{1i} \leq Z \right\}.$$

To compute $\rho_1^2(x)$, recall m_1 and m_2 . Also recall

$$f^*(u_1, u_2) = A(u_1)f(u_1, u_2)$$

$$f_1^*(u_1) = A(u_1)f_1(u_1).$$

With the same technique as in Lemma 3.2 we obtain, on $U_{1i} \leq Z_i$,

$$\begin{aligned} & \mathbb{E} \left\{ [(Y_{1i} - m_1(U_{1i}))m_2(x) - (Y_{2i} - m_2(U_{1i}))m_1(x)]^2 \mid U_{1i} = x, U_{2i} = v, U_{1i} \leq Z_i \right\} \\ &= \frac{(v - m(x))^2(1 - G^-(x + v))}{A^2(x)A^2(x + v)(1 - G^-(x))} = \frac{(v - m(x))^2}{A^3(x)A(x + v)}. \end{aligned}$$

Integrating out we obtain

$$\rho_1^2(x) = \frac{A^{-3}(x) \int \frac{(v-m(x))^2}{A(x+v)} f(x, v) dv}{f_1(x)}.$$

From (3.6) and (3.9) we therefore obtain

$$\begin{aligned} \rho^2(x) &= \frac{\rho_1^2(x) \int K^2(u) du}{m_2^4(x) f_1^*(x)} = \frac{A^3(x) \rho_1^2(x) \int K^2(u) du}{f_1(x)} \\ &= \frac{\int \frac{(v-m(x))^2}{A(x+v)} f(x, v) dv \int K^2(u) du}{f_1^2(x)}, \end{aligned}$$

as desired.

Next we study the bias. This is simpler since the two cases can be handled separately. If we apply standard expansions to (3.7) and (3.8) we get

$$(3.7) - (3.8) = h^2 B_1(x) + o_{\mathbb{P}}(h^2),$$

where

$$\begin{aligned} &B_1(x) \\ &= \frac{\int u^2 K(u) du \left\{ m_2(x) 2f_1^{*'}(x) m_1'(x) + m_2(x) f_1^*(x) m_1''(x) - m_1(x) 2f_1^{*'}(x) m_2'(x) - m_1(x) f_1^*(x) m_2''(x) \right\}}{2f_1^*(x)}. \end{aligned}$$

Some lengthy but tedious computations show that the term in brackets equals

$$A^{-1}(x) \{ 2f_1'(x) m'(x) + f_1(x) m''(x) \}.$$

Now use $f_1^* = Af_1$ and recall (3.6) and $m_2 = A^{-1}$ to get the formula for $B(x)$. This completes the proof of Theorem 1.1. ■

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